

COMPLEMENTARY SCALE, ASSURANCE, AND TRANSFER

Evidence that transfers across the architecture boundary.

Complementary experience

ZeusVu - Chief Pilot, Autonomous Systems and Aerial Data Operations | 2018-2022

- Led regulated drone operations across medical, energy, logistics, real-estate, and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.
- Designed medical drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders, and governed execution.

Cardinal Building Products - Regional Director | 2021-2023

- Built the company's first remote regional operation.
- Modernized distributed customer, CRM, commercial, fulfillment, logistics, digital marketing, SEO, and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.

Enviro Social Capital - Chairman of the Board | 2014-2017

- Co-founded and led a public-private impact initiative focused on sustainable infrastructure, green finance, and civic transformation.
- Helped shape and evaluate a \$200 million Social Impact Bond feasibility concept for Pittsburgh sewage and green-infrastructure improvement.

IEEE - CMPT and EDS Pittsburgh Chapter Chair | 2012-2016

- Led regional technology programming connecting industry, academia, STEM outreach, electronics, manufacturing, and emerging technology.
- Organized conferences, seminars, business tours, and technical programming.

Transfer map

AI capability design
Agentic AI, LLM workflows, RAG patterns, structured context, human review, escalation, decision records, evaluation, and governance rails.

Data and integration
ERP, CRM, telemetry, production, quality, support, purchasing, logistics, and engineering issue flows.

Reliability and assurance
Quality systems, root cause, process controls, regulated autonomy, safety, chain of custody, and high-consequence technical execution.

Enterprise adoption
Operating cadence, standard work, training, visibility, cross-functional translation, and distributed execution.

Education

University of Pittsburgh
Bachelor of Science
Academic grounding includes materials science, physical chemistry, and biology-related studies.

Credentials

Google AI Essentials
IBM Enterprise Design Thinking Practitioner
Six Sigma Certification
OSHA 10

Operating mechanisms I bring to the architecture role

Mechanism	Verified evidence	Architecture relevance
Operating visibility	Throughput visibility, dashboards, machine-fleet and customer signals, and performance measures.	Convert fragmented activity into a shared operating truth.
Decision cadence	Leadership routines, prioritized work, escalation paths, standard work, and accountability.	Move recurring architecture decisions out of improvisation without removing judgment.
Quality and learning	Quality systems, root-cause practices, process controls, and earlier risk detection.	Turn exceptions and incidents into control changes and reusable patterns.
Distributed execution	Remote regional operations plus CRM, commercial, fulfillment, logistics, and digital workflows.	Support federated teams with clearer interfaces, ownership, and adoption.
AI workflow design	Readiness frameworks, agentic patterns, human review, evaluation, governance, adoption, and reuse.	Attach AI to work, authority, evidence, economics, and sustainable ownership.
Architecture judgment Translate between outcome, data authority, AI behavior, human authority, control, integration, operations, economics, and reuse.		Scientific trust Preserve domain context while applying quality systems, traceability, regulated execution, and explicit decision boundaries.